ANUSHYANTH A

MECHANICAL DESIGN | MANUFACTURING | AEROSPACE MATERIALS

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PROFESSIONAL SUMMARY

M.Tech Manufacturing Technology student with hands-on experience in mechanical design, CAD/CAE, manufacturing, CNC/CAM, and aerospace materials. Experienced in CAD modelling, engineering drawings, CNC machining workflows, FEA-based analysis, and experimental research. Currently working on the repair of AA2219-T87 friction stir welds using Re-FSW, TIG, and Plug FSW techniques for aerospace applications at VSSC. Interested in aerospace mechanical design, manufacturing, materials, and engineering analysis.

EXPERIENCE

Kriatec Services Pvt. Ltd. March 2024 to July 2025

Application Engineer

- Developed **CAD models and detailed technical drawings** incorporating **GD&T** requirements for CAM applications.
- Simulated **turning and milling operations** using Hexagon Edgecam and developed post-processor models for CNC applications.
- Provided technical support to customers for CAM and CNC-related applications.
- Contributed to the development of technical courses for the company's e-learning platform.

Advanced Manufacturing Technology Development Centre (AMTDC), IIT Madras June 2026 to July 2026

Research Intern

- Designed and developed a vision-based measurement system for tool wear detection and dimensional measurement applications.

SKILLS

- Solidworks
- Creo Parametric
- Fusion 360
- Ansys Workbench
- Hexagon Edgecam
- Matlab
- GD&T
- CNC/CAM
- Abaqus

EDUCATION

Indian Institute of Space Science & Technology, Thiruvananthapuram 2025 to 2027

M.Tech - Manufacturing Technology

Sri Venkateswara College of Engineering, Chennai 2020 to 2024

B.E - Mechanical Engineering

PROJECTS

Repair Study on Friction Stir Welding of AA2219-T87 — VSSC 2026

Evaluating the mechanical and microstructural properties of defective AA2219-T87 friction stir welds repaired using Re-FSW, TIG, and Plug FSW techniques for aerospace applications. Investigating the effect of different repair approaches on tensile properties at room and cryogenic temperatures, microhardness, microstructures and fractographic features.

Design and Fabrication of an Electric Bike with Performance Optimization 2024

Designed and fabricated a lightweight, structurally stable, and aerodynamic bike chassis through multiple iterations of finite-element stress analysis, deformation simulations, and virtual crash testing in ANSYS. Evaluated battery performance under different driving conditions using MATLAB Simulink through trip-cycle analysis. Optimized the battery management approach based on the evaluated operating conditions.

Design and Fabrication of Quadcopter Drone 2023

Designed and fabricated a quadcopter drone focusing on stability, lightweight structure, and durability. Gained hands-on experience in aerodynamics, electronics, and structural design.

CERTIFICATIONS

Geometric Dimensioning & Tolerancing - Learnew (AICTE evaluated)

Fixture Design - Learnew (AICTE evaluated)

Automation in Manufacturing – NPTEL

ACHIEVEMENTS & LEADERSHIP

National E Bike Challenge 2024 **[Design Lead]**
Second Runner-up and Best Aesthetics Award

SIEP - E Bike Challenge 2023 **[Design Lead]**
Best Future Award

Render Royal - Inter college design competition
First Prize

Member of Bureau of Indian Standards

LANGUAGES

English - Professional

Tamil - Native